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RESEARCH ARTICLE

An exploration of psychological safety and conflict in first-year engineering student teams

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Abstract

Background: Developing teamwork skills is a central objective of engineering education. Psychological safety and conflict management are pivotal components of teamwork, yet despite their significance, research in engineering project-based learning (PBL) contexts is scant. Understanding students' experiences with psychological safety and its interaction with conflict is crucial to inform PBL pedagogy.

Purpose: This study delves into first-year engineering students' experiences of psychological safety and conflict, including their evolution in a PBL course.

Methods: Throughout the semester, we collected data from 82 students via written reflections and focus groups. Employing a thematic analysis underpinned by the team conflict dynamics model, we characterized students' experiences with psychological safety and conflict.

Findings: At the semester's outset, psychological safety was notably lacking. Students often hesitated to share ideas due to apprehensions about peer reactions and fears of negative judgments. As the semester advanced, consistent positive affirmations nurtured psychological safety, increasing students' confidence and readiness to discuss ideas openly and engage in healthy task conflict. Notably, process conflicts arising from absenteeism, poor communication, and procrastination were prevalent across teams. When unresolved, these conflicts eroded psychological safety, intensifying stress, exacerbating frustrations, and provoking relationship conflict.

Conclusions: Our study underscores the intertwined nature of psychological safety and conflict in shaping the first-year design experience in student teams. We urge faculty to recognize their pivotal role in fostering an inclusive culture and highlight pedagogical strategies that can bolster psychological safety at the onset, encourage healthy task conflict, and monitor unhealthy process and relationship conflicts.

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KEYWORDS

conflict, first-year engineering, project-based learning, psychological safety, qualitative research, teamwork

1 | INTRODUCTION

Effective teamwork and the capacity to establish a collaborative and inclusive environment are critical skills for engineering students, as established by the Accreditation Board for Engineering and Technology (ABET; ABET, n.d.). Engineering graduates have indicated that teamwork is one of the most important skills in their work (Passow, 2012). Teamwork skills are indispensable in modern engineering practice, where teams often grapple with complex, multifaceted problems with diverse, multidisciplinary team members. However, employers have reported that recent engineering graduates have not developed the interpersonal and professional skills to succeed in the collaborative engineering workplace (National Academy of Engineering, 2018). There has been a proliferation of active-learning pedagogical approaches in engineering education to help students develop teamwork and other professional skills that help meet workplace needs, including first-year cornerstone courses with design projects (Borrego et al., 2010; Froyd et al., 2012).

Engineering classrooms that use project-based learning (PBL) provide an ideal, low-stakes setting to foster teamwork skills and prepare students for the workforce. Positive team experiences in engineering education can be significant learning opportunities for students to practice developing teamwork skills. Yet, encounters with dysfunctional teams can induce negative perceptions of teamwork, potentially undermining the value of collaboration (Lingard & Barkataki, 2011). A concerning finding is when students feel they collaborate effectively, studies highlight there may still be a prevalence of inequitable teaming practices (Cross & Paretti, 2020; Hirshfield, 2018). Student teamwork experiences in engineering PBL can reinforce problematic teamwork habits and behaviors (e.g., task allocation grounded in gender stereotypes), which disproportionately affect historically marginalized groups in engineering and hamper broadening and participation efforts (Meadows & Sekaquaptewa, 2013; Tonso, 2006). Given these findings, faculty play a critical role in monitoring teams and supporting equitable teaming practices to promote positive student experiences and the development of teamwork skills.

Despite the need, engineering faculty have been found to have minimal training and can struggle with managing team dynamics (Paretti et al., 2013). A study by Beddoes and Panther (2018) reveals significant gaps in faculty practices across various engineering disciplines. They found that faculty often do not have clear rationales for team formation, lack mechanisms to identify and address team issues, do not actively oversee team roles, nor promote gender-inclusive teamwork practices (p. 338). Research suggests engineering students self-report improved teamwork and communication skills due to PBL courses (Beagon et al., 2019). However, these gains can be limited if the course does not explicitly address teamwork (Picard et al., 2022). These findings underscore the pressing need to enhance how engineering faculty can impart teamwork skills to students.

As discussed by Borrego et al. (2013), insights from industrial and organizational (I/O) psychology could prove invaluable for engineering educators navigating the complexity of team dynamics in the classroom. For instance, O'Neill and McLarnon's work (2018) characterizes the relationship between psychological safety, conflict, and team performance, providing a more nuanced framework to explore the complexity of team dynamics. Psychological safety refers to a shared belief held by team members that the team is safe for interpersonal risk-taking (Edmondson, 1999). The literature in various workplace settings and teams indicates many positive work outcomes from psychological safety, including performance, innovation, creativity, and learning (Newman et al., 2017). Google's People Analytics Unit even ranked psychological safety as the top trait of successful teams (Bergmann & Schaeppi, 2016). Jehn and Bendersky (2003) broadly define conflict as "perceived incompatibilities or discrepant views among the parties involved" (p. 189). The literature indicates, if managed effectively, conflict can catalyze learning and innovation (De Dreu & Beersma, 2005) but otherwise can be detrimental to team performance and individual well-being (De Dreu et al., 2001).

Research on psychological safety in engineering education is nascent, with most publications occurring in the last several years. Research on conflict in engineering education has approached conflict broadly, with efforts to identify the presence of conflict and conflict resolution in student teams (Novick et al., 2022) and characterize the impact of types of conflict and conflict management styles using correlational approaches (Sollitto & Mehrubeoglu, 2020; Strenger et al., 2020). However, there is little research aimed at gaining a deeper-level understanding of engineering students' conflict experiences and the processes through which group conflict emerges (Korsgaard et al., 2008), especially in

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relation to psychological safety. By understanding these interconnected elements and their impact on student learning in teams, engineering educators can scaffold the student experience more effectively by fostering more productive and psychologically safe team environments and managing conflict effectively.

Given these considerations, there is a need for empirical studies that delve into the nuanced experiences of engineering student teams' psychological safety and conflict to help us understand the impact on team dynamics and better inform future pedagogical initiatives. This study addressed this gap by exploring first-year engineering students' experiences with psychological safety and conflict in an engineering PBL course. In particular, we investigated the overall research question: What are students' experiences of psychological safety and conflict working in teams in a first-year engineering course? In addressing this overarching research question, we emphasized understanding the temporal evolution and interplay of psychological safety and conflict over the semester.

2 | LITERATURE REVIEW

This literature review delves into previous research on psychological safety and conflict across various fields while highlighting studies specific to engineering education.

2.1 | Psychological safety

Students are empowered to freely share ideas, take intellectual risks, and collaborate in a psychologically safe and inclusive environment, which can foster the development of well-rounded and creative engineers (Kazerounian & Foley, 2007). Research on psychological safety has been primarily conducted in organizational behavior, psychology, and management, focusing on business contexts. Research on psychological safety has found a range of benefits from enhancing individual and group learning (Bstieler & Hemmert, 2010; Ortega et al., 2014; Wong et al., 2010), boosting individual and team performance (Baer & Frese, 2003; Singh et al., 2013), fostering creativity (Carmeli et al., 2010) and work engagement (May et al., 2004; Nembhard & Edmondson, 2006), and including a sense of vitality toward their work (Kark & Carmeli, 2009) and knowledge-sharing (Mu & Gnyawali, 2003; Siemsen et al., 2009; Zhang et al., 2010). Psychological safety strongly correlates with innovation in research and development teams (Gu et al., 2013; Liu & Keller, 2021), manufacturing process innovation performance (Lee et al., 2011), and knowledge creation (Cauwelier et al., 2019; Kessel et al., 2012). Leadership has been identified as an antecedent that influences how psychological safety is shaped and leads to certain outcomes, such as the capacity for creative problem-solving (Carmeli et al., 2010), suggesting faculty influence psychological safety in student teams in engineering classrooms.

Research on psychological safety in engineering education contexts has just begun to emerge in recent years. Recent research reveals a disparity in psychological safety in educational settings, showing White students generally report higher levels of safety and women experience notably less psychological safety compared with men (Ohland et al., 2024). Further, early research has found psychological safety to significantly predict conflict and cohesion in engineering student teams at the individual and team levels (Beigpourian et al., 2019). Recent studies have investigated psychological safety in engineering student teams over time (Cole, Marhefka, et al., 2022; Cole, O'Connell, et al., 2022). Cole, Marhefka, et al. (2022) and Cole, O'Connell, et al. (2022) found that psychological safety was associated with the number of viable ideas generated by first-year engineering student teams. In engineering first-year design teams, communication, coordination, cooperation, composition, conflict, creativity, and cohesiveness were found to influence psychological safety (Cole, Marhefka, et al., 2022; Cole, O'Connell, et al., 2022). Findings from Cole, Marhefka, et al. (2022) and Cole, O'Connell, et al. (2022) mixed methods study on psychological safety in engineering student teams highlight its relationship with conflict, noting conflict is a double-edged sword in its impact on psychological safety, meaning conflicts often viewed negatively, and can potentially enhance psychological safety and overall team performance when managed well.

2.2 | Conflict

Similar to research on psychological safety, conflict and conflict management have been studied extensively in psychology, management, and organizational behavior. Jehn's conflict typology has been a common approach in the

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literature to characterize the impact of task, relationship, and process conflict on outcomes such as team effectiveness and performance (e.g., Jehn et al., 2008; Jehn & Mannix, 2001). Research results on the positive and negative impacts of conflict type on team performance are multifaceted. Task conflict promotes discussion of different ideas and can be beneficial to teams in fostering creativity and innovation (Weingart et al., 2010). However, studies highlight the complexity of task conflict, indicating its effects can be positive and negative and likely influenced by various mediating factors (Puck & Pregernig, 2014). Both process and relationship conflict more consistently lead to adverse outcomes for teams. Process conflict tends to be a symptom of a team's inability to allocate workload, roles and responsibilities, and timelines fairly (Jehn et al., 2008). Thus, teams must identify and manage these process conflicts in the early stages of team development before they become problematic and detrimental to team performance (Jehn & Mannix, 2001). Relationship conflict invokes feelings such as anxiety, anger, and fear (Jehn, 1995), which leads to less mental resources dedicated to flexible thinking and considering other viewpoints (Behfar et al., 2008). Consequences of poorly handling these types of conflict in teams generally reduce individual and team performance and effectiveness (West & Hirst, 2005) and have psychological impacts, such as negative emotions, demoralization, and burnout (Bloom, 2010).

We highlight engineering education research studies involving conflict typology, specifically task, relationship, and process conflict. Strenger et al. (2020) found process conflict had the highest influence on team member effectiveness in engineering teams, although task and process conflict negatively impact team effectiveness. Ohland et al. (2024) found differences in conflict among student demographic groups. For example, male students reported higher task conflict. Their work found conflict to be the most significant factor influencing team effectiveness compared with psychological safety and cohesiveness. However, the authors acknowledge the challenges of relying solely on quantitative measures, where many students favorably rate their teammates on these scales (Ohland et al., 2024; Wei & Ohland, 2021). This phenomenon in peer evaluation ratings is referred to as leniency bias and is influenced by friendships and personality perceptions (Mayfield & Tombaugh, 2019). Managing conflict has been a consistent topic of interest and recognized need, with multiple studies exploring conflict management training and interventions to improve outcomes for student teams (Magana et al., 2022; Novick et al., 2022; Pertegal-Felices et al., 2019; Ryan et al., 2023; Usprech & Lam, 2020).

2.3 | Understanding the role of psychological safety and conflict in engineering education

These findings collectively support the vital role of psychological safety and conflict in teams. Considering the dearth of research available on engineering students' perceptions of psychological safety, further exploration of how students experience psychological safety in engineering teams and having a more nuanced understanding of its relationship with conflict is critical. A better understanding of how psychological safety develops and influences conflict can inform pedagogy and awareness, so engineering instructors can monitor and support it more effectively and improve teamwork outcomes and overall team experiences. Notably, although substantial research on team dynamics exists in engineering education, a significant gap remains that this work will begin to address: the temporal relationship between psychological safety and conflict has seldom been explored qualitatively, especially in the context of student–team experiences over time.

2.4 | Theoretical framework

The theoretical foundation for our study is grounded in the team conflict dynamics model (TCDM) proposed by O'Neill and McLarnon (2018). This model was developed based on the findings of meta-analyses in the field of I/O psychology to provide an in-depth perspective on how psychological safety interacts with team conflict dynamics to influence team performance. This comprehensive model explores the multifaceted nature of team conflict by focusing on the conflict states (i.e., task, relationship, process) and conflict management processes in the team conflict system (see Figure 1). Conflict transformations are considered over time by examining the interplay between psychological safety, team conflict, and team performance feedback as elements that reciprocally affect one another in a feedback loop. The model also incorporates team design and organizational environment as contextual features that influence the interplay of the elements in the model.

A significant aspect of the TCDM is the concept of conflict state profiles, characterized by different combinations and levels of task, relationship, and process conflict. Task conflict refers to disagreements about the content and outcomes of how the team should perform a task. Process conflict refers to disagreements about the logistics of task

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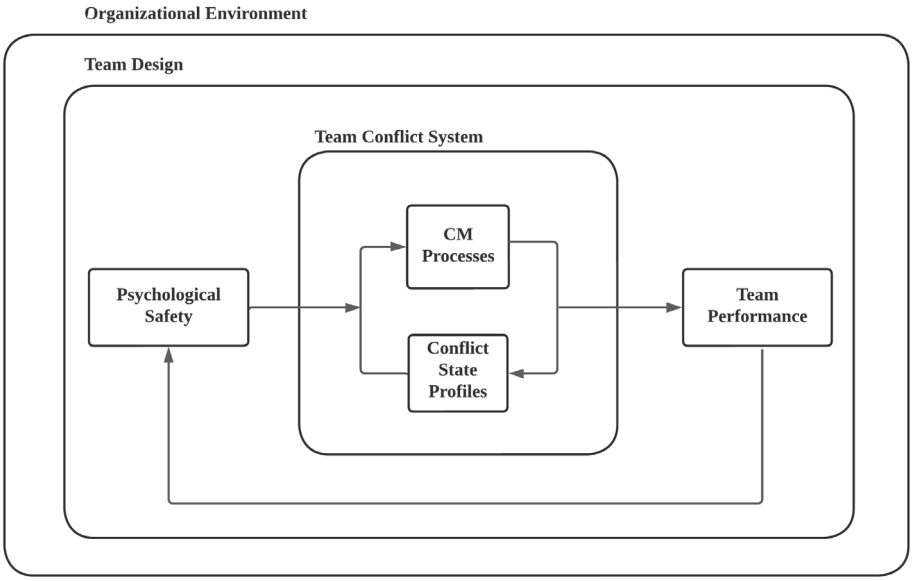


FIGURE 1 Team conflict dynamics model (O'Neill & McLarnon, 2018). CM Processes refer to conflict management processes.

accomplishment. Relationship conflict refers to interpersonal incompatibilities, tension, and animosity among team members. We expected these different types of conflict interact closely with psychological safety. These profiles offer a nuanced perspective on team conflict as a dynamic, evolving state characterized as healthy or unhealthy. For instance, a team could experience a task-conflict-dominant profile where disagreements about task content are moderate to high, but relationship and process conflicts are relatively low. This would be considered a healthy conflict profile as task conflict in teams can positively support group performance if actively managed (De Dreu & Weingart, 2003; DeChurch & Marks, 2001). In unhealthy conflict profiles, process and relationship conflicts are more likely to be at moderate to high levels. Indeed, the literature suggests process and relationship conflict negatively affects group performance (Behfar et al., 2011; De Dreu & Weingart, 2003). Depending on the team's context and evolution over time, a team could experience a range of these conflict profiles. Schoen et al. (2024) have also expanded upon the TCDM by incorporating ways creative personalities in teams impact the team conflict profile and subsequent team outcomes. Thus, it is important to also take into consideration numerous factors that can impact team performance, which can be related to psychological safety, in addition to a team's creative personalities and openness to experiences, such as voice disparity (Brykman & O'Neill, 2021), team learning activities (Kostopoulos & Bozionelos, 2011), and team diversity (Ensari et al., 2015).

Our study leveraged the TCDM as a guiding theoretical framework, allowing us to understand the complex and dynamic nature of psychological safety and conflict in PBL environments. By considering temporal transitions of conflict profiles, we aimed to provide meaningful insights to inform how instructors can support their student teams in adopting effective conflict management strategies that lead to healthy conflict profiles in engineering education PBL contexts.

3 | METHODS

3.1 | Participants

We recruited participants for this study from a first-year foundational engineering course, the second part of a required two-course sequence. Class sizes typically range from 60 to 72 students. The first author taught the off-cycle, second-semester foundations course during the Fall 2022 semester at a large, R1 research-intensive university in the mid-Atlantic region of the United States. Across two different course sections, 82 students chose to participate in the study, and 21 students opted out of the study. Thus, approximately 80% of the class participated in the research study. Demographic distribution of student participants included predominantly men, with women and non-binary students making up approximately 20% of the population, as detailed in Table 1. We note that individuals were able to identify with multiple races or ethnicities.

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TABLE 1 Demographics of participants.

	Number of students
Gender	
Male	59
Female	15
Non-binary	1
Prefer not to say/no answer	7
Race or ethnicity	
Asian	30
Black or African American	3
Hispanic, Latino, or Spanish Origin	8
Middle Eastern or North African	6
Native Hawaiian or other Pacific Islander	1
White	35
Other or prefer not to say/no answer	8

We recruited participants in class. Students were allowed to self-select to participate in the study. Importantly, enrollment in the study was voluntary, and we informed students their decision to participate would not impact their course grades or standing. However, all students were still required to complete reflection assignments and participate in focus groups, as we graded these assignments as part of the class. We did not include any data from nonconsenting students in the study.

3.2 | Course context

We situated this study in a required, PBL first-year engineering course. This course introduces students to foundational engineering concepts through a semester-long design project. In this PBL course, teams with 4–5 students collaborated to design and develop a solution that addresses a team-selected problem or need. Teams were required to produce a physical prototype, a 3D model using computer-aided design (CAD), and program a microcontroller using multiple sensors and actuators to collect, process, and output data. The primary learning objectives of the course were to develop students' proficiency in applying the engineering design process, having the ability to use engineering tools, and using problem-solving, teamwork, and communication skills.

In alignment with recommended practices (Oakley et al., 2004), CATME, an instructor tool to support team formation and monitoring, was used to optimize the team formation process. There was a particular emphasis on not siloing women and minorities on teams and diversifying teams based on skill set. The instructor facilitated several team composition and norming activities in the early stages of the course intended to support psychological safety and effective conflict management, as described in Table 2.

Early in the project, teams navigated a 7-week project scoping phase, where students engaged in various activities and completed individual and team sub-deliverables. This phase also emphasized technical skills training on CAD modeling, circuits, and Arduino programming, and individual assignments for students to practice further and develop these skills. These training sessions gave students foundational technical skills to contribute to their final prototype. This initial scoping phase culminated in a Design Review, where teams presented their project proposals and received extensive formal feedback. Following this, in Weeks 8–11, teams were divided into subteams, focusing on physical prototyping, programming, and 3D modeling. This prototype planning phase concluded with a Manager Review in Week 11, providing a checkpoint for the instructor team to ensure teams were on track and the proposed projects aligned with course requirements. After this, teams concentrated on prototyping and their final project deliverables, culminating with in-class presentations during Week 14, which included demonstrations of the teams' final prototypes. Figure 2 summarizes the project phases, key checkpoints, and timing of data collection.

We asked students to complete CATME self and peer evaluations after each key checkpoint (i.e., Design Review, Manager Review, and Final Project Demo). Prior work has demonstrated that using peer evaluations provides an

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TABLE 2 Team composition and norming interventions.

Intervention	Type	Description
CATME team formation survey	Individual assignment	Students were assigned to teams by the instructor based on individual characteristics (e.g., gender, race/ethnicity, skill set).
Intro to psychological safety	Activity and interactive lecture	Students participated in an activity that introduces them to psychological safety. A case study was reviewed and discussed in class.
Team asset mapping	Activity	Students participated in an icebreaker activity in which they shared their unique strengths, interests, and relevant previous experiences.
Team contract	In-class activity and team assignment	Students drafted a list of team expectations or norms on communicating, delegating tasks, evaluating work quality, conducting meetings, managing conflict, and so forth.
Conflict management workshop	Interactive lecture and class activity	Students learned about different types of conflict and conflict management approaches and participated in a scenario-based, role-playing activity where they discussed a relevant conflict scenario.

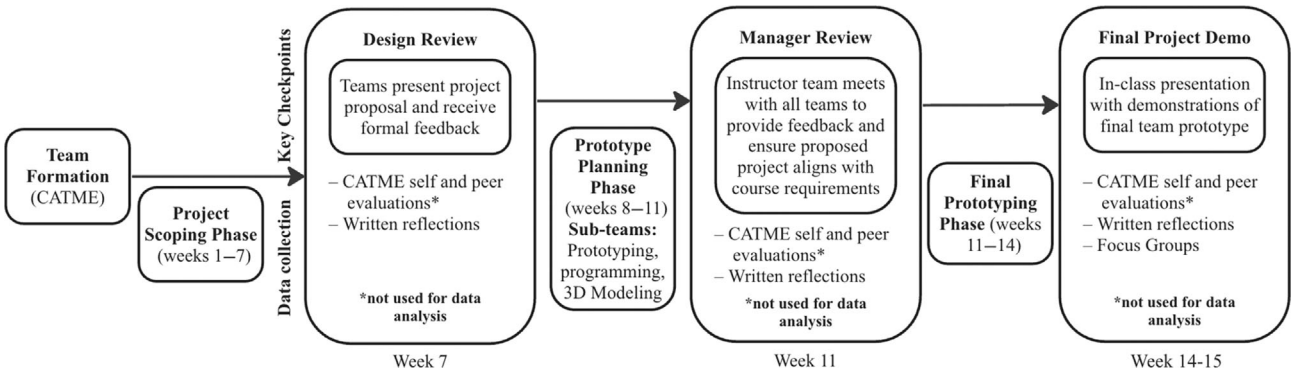


FIGURE 2 Course context overview and key checkpoints.

accountability mechanism that helps mitigate social loafing (Brooks & Ammons, 2003; Hall & Buzwell, 2013; Harkins & Jackson, 1985; Price et al., 2006), a frequently cited negative behavior in engineering education in which some students exert less effort in group projects than if they were working individually (Borrego et al., 2013). This approach offered periodic insight into team dynamics throughout the course, enabling the instructor to identify and address significant conflicts. The instructor screened and released peer ratings and peer comments to students as a formative feedback mechanism for the first two peer evaluations the following week after students completed the evaluations.

3.3 | Data collection

In the data collection and analysis steps for this study, a robust quality management process was implemented with purposeful considerations for validation and reliability, as Walther et al. (2013) recommended. These considerations are highlighted and discussed throughout these subsections.

We collected student data through written reflections and end-of-semester focus groups. Written reflections are a typical component of the pedagogical structure in PBL courses (Mills & Treagust, 2003) as this activity requires students to reflect on their experiences critically, an exercise recognized as a valuable part of learning in engineering education (Turns et al., 2014). Collecting data through reflections provided an established and effective method to gather students' perspectives on their teamwork experiences while supporting their development of teamwork skills.

Students completed a total of three reflections after the project scoping phase (Week 7), initial prototyping phase (Week 11), and after the project (Week 15). Having students complete reflections throughout the semester encouraged continuous monitoring and attention to their individual development of teamwork skills and how their team dynamics were evolving during the course. All reflections had a prompt with multiple questions, encouraging students to answer in a short paragraph. Not all reflections directly asked students about psychological safety and conflict; however, the

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questions asked students about disagreements, collaborations, and communication, potential antecedents to psychological safety and conflict. For example, one of the reflection questions asked:

> In a paragraph (4–6 sentences), describe a time your group had to make an important or difficult project-related decision so far this semester. Were all of the perspectives of the group members considered? Why or why not? Were there any disagreements or conflicting opinions? If so, how were the disagreements among team members handled? Are there ways the team could improve in how it makes important team decisions?

The purpose of this reflection prompt example was to uncover ways student teams experienced task conflict without explicitly mentioning task conflict. We include the reflection prompts in Appendix A.

We instructed students to answer the prompts in the situation, task, action, and result (STAR) format, a common interview response technique. The STAR format required that students describe the situation, the task, action, and result of the situation. By requiring students to answer our reflection prompts in this specific format, they are describing a specific experience that may be focused on or closely related to team conflict or psychological safety, the phenomena of interest for this study. The use of the STAR format was intentional to help align student reflections with addressing our overarching research question.

In addition to written reflections, we conducted focus group discussions to offset the potential limitations of relying on formal class reflections for data collection. Students may not be as transparent or authentic in their class reflections during the semester for various reasons. For example, students may want to portray themselves more positively for grading purposes or avoid disclosing certain information because of concerns that may negatively impact a teammate or their current team dynamic. Focus groups also have their benefits as they allowed us to capture nuances of individual experiences in a communal context and enable students to build off the experiences of others, echoing the active, social learning characteristics of PBL (Prince, 2004). Focus group discussions also served as a debriefing activity for students, supporting their learning experience by exposing them to various perspectives and team experiences.

In the final reflection, students reflected on their semester and collective team experience. We conducted the focus groups after students completed the final reflection and semester project. Several focus group questions aligned with the final reflection, so students already had a chance to reflect on their semester team experience before discussing it in the focus group. Focus group questions were open-ended, asking students to think about how their team dynamics evolved, what conflict their teams experienced over the semester, and which specific learning takeaways and skills they will take moving forward. Students were assigned to focus groups with three to four students per group and one to two facilitators (i.e., the authors of this paper, excluding the author who served as the course instructor), which lasted about 25 min. Nonconsenting students were separated into distinct focus groups to maximize data collection. We analyzed 19 focus groups with consenting students for this study. We included the focus group questions in Appendix A.

The first author (i.e., course instructor) did not facilitate any focus groups to ensure students felt comfortable sharing their honest thoughts about their classroom teamwork experiences. The focus group assignment was random but ensured students were not in the same group as anyone on their semester-long team to encourage authentic conversations about their experience. Additionally, we intentionally assigned at least two women to each focus group, so they were not isolated. These considerations in the design of the focus groups were intentional in enabling students to feel more comfortable discussing their team experiences and help mitigate potential limitations of using reflections as our only data source.

3.4 | Data analysis

In our study, we intended to understand the lived experiences of our participants through a phenomenological approach. However, a thematic analysis was more suited to our findings due to tensions in using an interpretive phenomenological analysis with larger sample size, limited depth of the data, and more collaborative approach to data analysis (Kirn et al., 2019). The decision to make this adjustment aligns with our interpretivist philosophical stance on our analysis, adapting our methods in response to the nature of the data we collected. We conducted a thematic analysis incorporating both deductive and inductive approaches using steps outlined by Braun and Clarke (2006). We used the TCDM to interpret and understand students' experiences with psychological safety and conflict, as captured in their written reflections and focus group discussions. Thus, our analysis began with a deductive focus on our primary phenomena of interest: psychological safety, task conflict, process conflict, relationship conflict, and conflict management. In these constructs, we inductively identified patterns and trends, concentrating on students' experiences regarding psychological safety and conflicts over the semester.

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In the initial stage of our analysis, we immersed ourselves in the reflection data. Two members of the research team and first author reviewed all the data and identified excerpts that illuminated participants' experiences of psychological safety and conflict. The primary output of this step was identifying and organizing the data to further our understanding of how students experienced psychological safety and conflict in alignment with the TCDM.

Following clustering and thematizing of the data, the first author examined the clustered data and generated an initial set of codes for psychological safety and conflict types. We conducted several rounds of axial coding to consolidate codes into emergent trends and patterns. Given the large participant pool ($n = 82$), we approached the interpretation of experiences collectively rather than attempting to link individual reflections to focus group transcripts. We developed a preliminary codebook from this process that captured how students experienced psychological safety and different types of conflict.

In alignment with the theoretical framework, we created an additional theme for performance feedback to capture the team performance feedback loop in the TCDM. Performance feedback was not a focus of this study as it is strongly intertwined with instructor practices and is not necessarily generalizable. However, several notable examples in our data illustrate its relationship to psychological safety and conflict. We determined that briefly discussing performance feedback in our findings would be beneficial. We captured specific approaches to how students managed each type of conflict under each conflict type (e.g., process conflict). We used the conflict management theme to summarize overall patterns or trends in how students managed all types of conflict.

Our initial codebook, complete with brief descriptions and representative examples, was reviewed by the research team. The first author refined the codebook based on the feedback. Next, the first author did the first review of the focus group data. In the second review, the first author conducted a round of deductive coding using the codebook developed from the reflection data. They applied open coding for relevant data not present in the previously developed codebook to the study. The focus group data confirmed the reflection data; however, these data provided a more detailed, granular descriptions of the development of psychological safety and its interplay with conflict. Additionally, we uncovered new insights into students' perceptions of task conflict that had not already been captured in the reflection data. Triangulation across different data sources supports the reliability of the results.

Following the final stage of the data analysis, the first author proceeded to craft detailed descriptions of each theme, integrating supporting quotes from the data. In addition, they integrated insights into the interplay between psychological safety and conflict into the thematic descriptions. Our analysis culminated in an integrated and comprehensive understanding of students' experiences with psychological safety and conflict organized deductively in the team conflicts dynamics model. The research team played an integral role in reviewing and refining the composite description in the findings section, ensuring its clarity, conciseness, and representation of our data. To ensure reflexivity, our research team engaged in regular meetings to critically assess and discuss the analysis and findings from the data throughout the process. These discussions helped identify and address potential biases and strengthened the trustworthiness of our overall outcomes.

3.5 | Positionality

The authors of this study bring diverse experiences and insights related to conflict management and involvement in first-year PBL courses. The lead author had previously taught over 10 sections of first-year engineering PBL courses across two institutions over 3 years, deepening his appreciation for psychological safety and effective conflict management in this context. Consequently, he has initiated integration of content and training to nurture these competencies in students. His personal beliefs and aspiration to comprehend students' experiences with these elements, so he and other faculty can more effectively address them, was a primary impetus for this study. He thus strived to authentically capture and represent student perceptions throughout this study.

To counterbalance potential biases, co-authors played pivotal roles in facilitating focus groups, categorizing the data, establishing thematic relevance to the phenomena in question, and critiquing numerous iterations of the findings. Among this study's five authors, three authors have a distinct combination of personal and professional experiences that directly inform our perspective and approach to this research. For the women authors who have navigated careers in diverse sectors of the engineering industry, including aerospace, electrical, and biomedical engineering, they have experienced the realities of team dynamics in academia and professional arena working in diverse, multidisciplinary, multicultural teams. They have seen how these dynamics, initially shaped in the classroom, extend into the workforce, where these dynamics can perpetuate biases, reinforce stereotypes, and generate noninclusive and problematic work environments. From these experiences, they believe an urgent need exists to more explicitly educate engineering students and instructors about the value of working on diverse teams and fostering inclusive, psychologically safe teams through effective conflict management. Implementing such initiatives can stimulate a significant shift toward a more inclusive engineering workforce.

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Moreover, as all five authors are former engineering students and present educators, they have witnessed the profound impact group work can have on students' perspectives and future outlooks of the engineering field. We approached this research rooted in the belief students possess the potential to learn and cultivate self-awareness and teamwork skills through their engineering education experiences with effective training and mentorship. All five authors' collective research group experiences, beliefs, and academic and pedagogical insights have shaped our approach to this work.

4 | FINDINGS

Our findings are shedding light on how students experience psychological safety and different types of conflict (i.e., task, relationship, process) in a first-year engineering PBL course's team-based context. Data analysis followed the TCDM, and findings are themed accordingly: (a) psychological safety, (b) team conflict state profiles (i.e., task, process, and relationship conflict), (c) conflict management processes, and (d) performance feedback. Figure 3 highlights the primary codes under these themes. Appendix B contains a codebook with descriptions and additional representative quotes for psychological safety and conflict codes. Patterns that emerged illustrate a clear relationship between the constructs in the TCDM. In our findings, we assigned a pseudonym for quotes from student reflections, but we did not connect pseudonyms to focus group quotes because we did not want to make assumptions about who was speaking.

4.1 | Psychological safety

4.1.1 | Improvements in psychological safety and open communication

Our collective data indicate many students experience notable improvements in psychological safety over the semester, which impacted their ability to openly communicate. Students often described early interactions as uncomfortable or

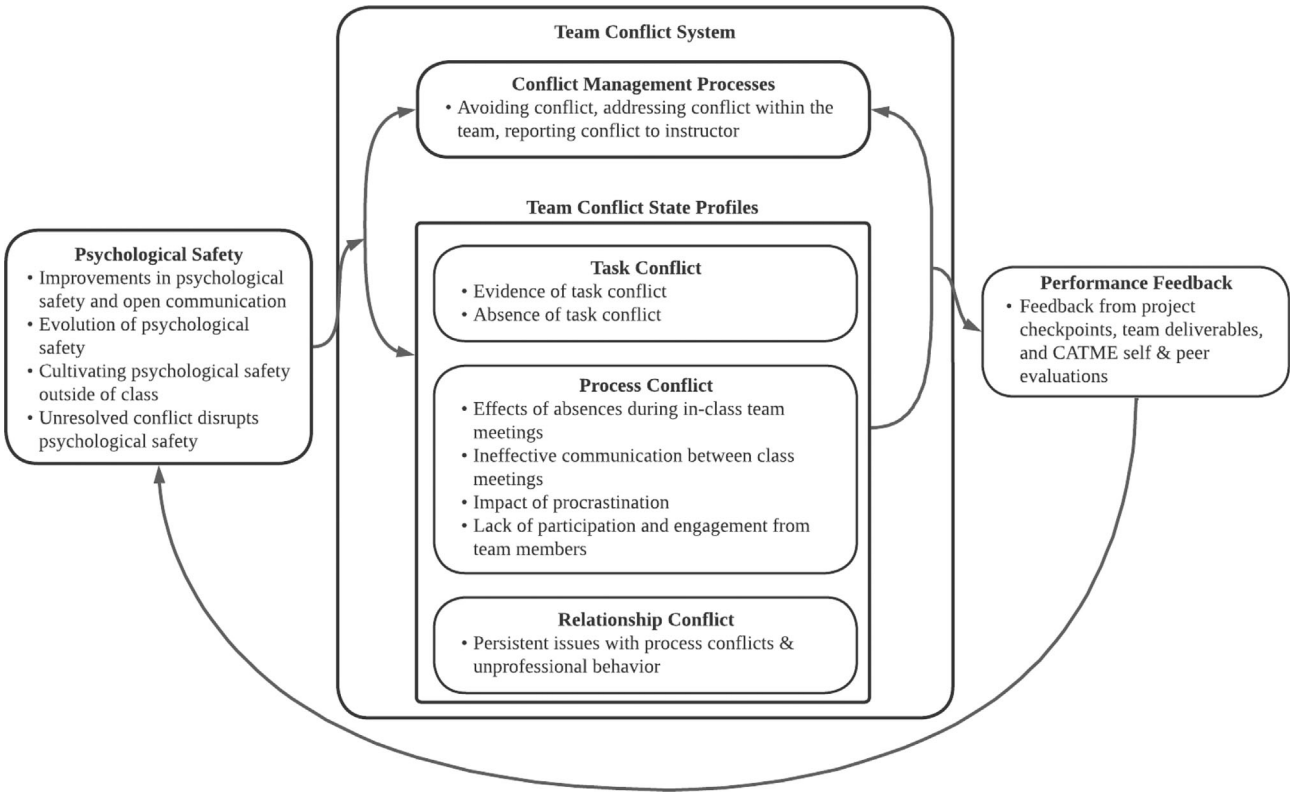


FIGURE 3 Overview of codes (in bullet points) of psychological safety and conflict organized in the TCDM to demonstrate the dynamic interaction between our themes and codes. We removed organizational and team design contextual factors to simplify the diagram.

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awkward, with students being tentative to share ideas due to uncertainty about how their peers might respond. As the course progressed, many students felt more comfortable openly voicing their opinions and ideas without fear of negative judgment. This sentiment is captured well by a student during a focus group:

> I think the biggest change for our group was comfort. Just people being okay with sharing and being okay with even having their ideas shot down or whatever. Being okay to come up with those and get feedback. Because in the beginning, it's like you don't want to share something and have four people you don't know be like, "No, it's dumb. We don't want to do that." But once you get comfortable with people and you can see their personalities, you're more comfortable with that. I definitely saw that evolve over time.

Psychological safety was still being developed early in the semester, impacting students' willingness to share ideas. Students describe cautiously approaching their initial interactions with teams and hesitating to openly communicate criticisms, as evidenced by a student in a focus group:

> I think one of the biggest things that have changed over the semester were the level of comfort with each other because I'm personally [a] quiet person, so I don't talk much with other people, so at first, I was a bit shy and I didn't really vocalize my ideas. I told them the good ideas, ideas that I thought were kind of good, but the ideas that I had doubts on, I just held in.

These two quotes highlight how students' comfort levels early in the semester influenced their willingness to share their ideas. Furthermore, in these early interactions, positive affirmation and an open reception of ideas were critical to nurturing a psychologically safe environment. Consistent positive interactions increased students' willingness to share ideas, bolstering the teams' psychological safety. This is described well in Chin's second reflection:

> I was a bit quiet in the beginning stages of the project, but I suggested a few ideas to my team which they seemed to really like. As a result, I felt encouraged to share more of my ideas and feedback with my team members which they have taken with a positive attitude and are appreciative of. This really motivated me to be more vocally active and participate more in discussions as my opinion matters to the group.

Positive feedback from teammates did not necessarily mean always agreeing with all the ideas shared, but being more respectful and listening to others with an open mind:

> Other times, my teammates have given me respectful criticism when necessary. At first, I was nowhere near as comfortable sharing my ideas with my group, but as the semester went on I grew more and more accustomed to communicating with them. (Jake, Reflection 2)

Alternatively, some students described feeling comfortable sharing their thoughts with their group from the very beginning, "I started the semester feeling like I could share my ideas without fear of judgment, and that hasn't changed" (Kenji, Reflection 2). Several of these confident team members described their role in catalyzing group conversations early in the semester by sharing their thoughts first and then encouraging other team members, especially socially reserved team members, to then provide input:

> My team is fairly introverted and needs some nudges to get conversations going. I will normally start talking first just to get the ball rolling and ask if anyone else wants to start. If no one will start, I will normally go first and then hand it off to someone else. I think my team members feel comfortable sharing their thoughts, it's just a matter of who is going to go first and who is going to break the ice to get things started. (Mei, Reflection 3)

Having a student willing to be vulnerable and break the ice in early group conversations can create more comfort for others to follow suit and openly share their ideas. This approach was specifically referenced as helpful at the beginning of the semester; however, as the collective psychological safety of the group increases through positive interactions, these groups may have decreased their reliance on one team member to always initiate the conversation. We did not have any data that suggest this approach was used in the middle or latter parts of the semester.

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In addition to idea-sharing, establishing psychological safety, often described by students as feeling more comfortable with teammates, appeared to influence help-seeking behaviors. This is illustrated by a student in a focus group comment:

> But once we got comfortable communicating and stuff, we had no issues, for example, asking for help. Because there's going to be a time where you just can't do stuff on your own. Asking for help is not going to be a big deal.

Our data strongly suggest it is a common experience, but not necessarily a uniform experience, for students to feel some level of discomfort in their early team interactions, which impacts their willingness to communicate openly. Positive, affirming team interactions are critical in a team's early stages to bolster student psychological safety, thereby promoting idea-sharing and help-seeking.

4.1.2 | Evolution of psychological safety

How teams cultivated a strong sense of psychological safety appears to have varied from team to team. The process of establishing psychological safety could occur quickly in the first few group interactions or more gradually over the course of weeks. For example, one student in a focus group shared their experience of quickly bonding with their group members:

> We were able to find some topics that were outside of class, that we were able to bond over, and then that just broke the ice from day one. And then from there, we were able to work pretty well with each other and be pretty comfortable [with] how we talked to each other.

Students connecting outside of class appeared to greatly accelerate psychological safety; this will be discussed further in the next theme. Most students discuss psychological safety evolving in their teams gradually and primarily over the first half of the semester. This finding was clearly evident in latter student reflections and focus groups when students retrospectively discussed how their team dynamic evolved. For example, Raj described a clear evolution of their team dynamic in their third reflection:

> I believe that the team dynamics changed significantly from the beginning to the end of the semester. In the beginning, we were shy from each other; therefore, we did not share a lot of conversation. In the middle, we got used to each other, and we were able to talk and discuss ideas. In the end, we reached the point where we understood each other very quickly and we respected and liked each other.

The overwhelming sentiment among students is there is an initial awkward phase with certain social boundaries teams need to overcome to establish psychological safety. By the time of our first reflection (Week 7), many students generally described feeling comfortable and also noted their teammates appeared to feel more comfortable sharing their thoughts and ideas in group interactions. By the end of the semester, students noted improvements in open communication from increased comfort led to further understanding and enhanced collaboration. This is epitomized in Liam's third reflection:

> The main change in our team dynamics was the ease of communication and feeling more comfortable communicating with one another as the semester went on. In the beginning of the semester as we were being assigned our teams, it was of course a little awkward. By the middle, we began to become a much better team by communicating more frequently, and by the end we were operating at a high level and collaborating frequently to complete all project deliverables.

Cultivation of psychological safety early in the semester is a particularly noteworthy phenomenon as many students retrospectively describe their comfort levels improving. Most students appear to be comfortable in their teams by mid-semester, which positively enhances communication and collaboration gradually for the remainder of the semester. By the end of the semester, students noted a stark contrast in their comfort, communication levels, and collaboration.

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4.1.3 | Cultivating psychological safety outside of class

Meeting outside of class in informal settings to complete group work together was noted as a critical component of establishing psychological safety. This phenomenon is captured well by Omar in their first reflection:

> I personally feel comfortable sharing my ideas which I'm very happy about because I'm a little bit shy but I'm pretty comfortable with my team. I always see my teammates speak when they have something to share so I think everyone is comfortable around each other. Additionally, some of us hung out with each other outside of the classroom, and I think that has strengthened the friendship among us and that makes it a lot easier to work together.

Meeting outside of class accelerated the process of students getting to know one another and becoming more friendly with each other, accelerating the process of cultivating psychological safety. Notably, some students describe this increased friendliness as beneficial to the team, especially in their communication. An example of this friendliness can be seen in the focus group excerpt:

> It kind of evolved to more of a friendship. So, I really enjoyed that kind of relationship change between my team. I just felt like it just made things so much more easier to communicate because we just felt closer to each other. We bonded over things that weren't just this class.

Although friendliness among team members was generally perceived as a positive shift for groups, one student does note one drawback of digression with the team sometimes getting distracted with off-topic discussions during project meetings. Our data suggest ability to develop friendships outside of class helps promote psychological safety among teams. However, it may hinder team effectiveness if it distracts teams during meetings.

4.1.4 | Unresolved conflict disrupts psychological safety

Although our data suggest teams generally cultivated psychological safety in the first half of the semester, enhancing collaboration for the remainder of the project, there is evidence some teams struggled to maintain this psychological safety as conflicts emerged. Persistent, unresolved conflicts, especially toward the semester's end, were reported to negatively impact team dynamics, including psychological safety. These conflicts, including connections to psychological safety, are discussed in the subsequent themes on task, process, and relationship conflict. Figure 4 provides an overview of the temporal evolution of psychological safety and conflict.

4.2 | Task conflict

4.2.1 | Evidence of task conflict

There was strong evidence of task conflict being present; however, it is not always perceived as a conflict, but rather a difference in opinions or ideas. This perception is succinctly described by a student in a focus group:

> I wouldn't really call it conflict, but obviously through prototyping people would have a difference in opinion. It did take a bit to come up with a solid design.

Task conflict was evident throughout the various stages of the design process, especially the project scoping and prototyping phases, because teams had to make pivotal decisions in scoping their projects and devising a design for their prototypes. The process involved discussing different ideas, often pertaining to the design features, and grappling with varied technical solutions, sometimes requiring additional research. Our data suggest task conflict was effectively managed by most teams, who were generally able to reach a consensus on decisions without major conflict. A student describes this practice of their team consistently managing task conflict and making team decisions in a focus group:

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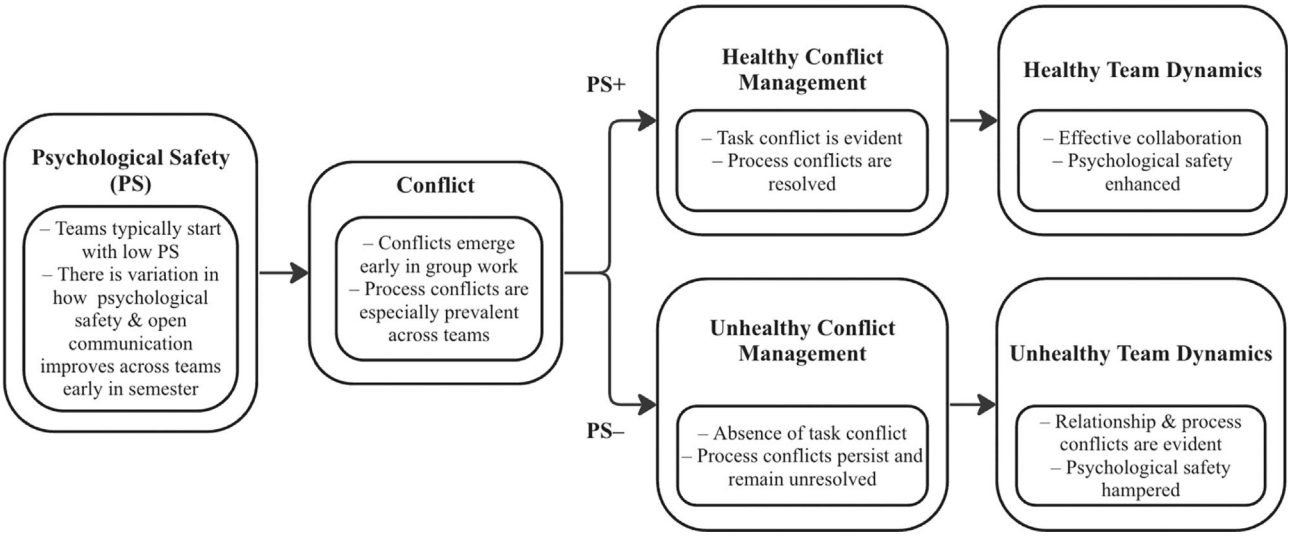


FIGURE 4 A temporal overview of how the evolution of psychological safety and conflict can promote healthy or unhealthy team dynamics in our first-year PBL context. Improvements in psychological safety (PS+) support healthy conflict management and team dynamics, and a lack of substantial improvement in psychological safety (PS−) can promote unhealthy conflict management and team dynamics.

> For our team, there weren't really any major conflicts. There was just a ton of smaller conflicts throughout the semester. With our project, there were just a lot of design decisions that had to be made. Essentially, every single class, every single time we met, we were having to constantly make multiple designs on our decisions on sensors or what we're going to do.

Teams effectively managed task conflict by fostering input, practicing active listening, offering respectful criticisms, and making group decisions. This process of open communication and valuing one another's input aiding the problem-solving process is captured well in Ryan's second reflection:

> My comfort level within this group has only gotten better as time went on. We all are very comfortable with sharing our own thoughts and we make everyone feel respected. There isn't a time where we take someone's ideas for granted or shut them out immediately. Any ideas that I share or that any other member shares are all taken into consideration and every person has a say on what the group decides.

Psychological safety was thus critical to enabling healthy task conflict as it impacted students' willingness to openly share and discuss ideas candidly, respectfully, and professionally. Disagreements are handled respectfully as indicated by Ben in their first reflection:

> We all treat each other with respect and avoid getting too combative with one another over differences in opinions. I feel extremely comfortable sharing ideas during team interactions. I believe that my teammates also feel extremely comfortable sharing their ideas during team interactions. We never run out of ideas.

Many students discussed their teams successfully amalgamating the best aspects of different ideas or individual designs to create a better overall design, suggesting many teams reaped the benefits of healthy task conflict supported by open communication strengthened by psychological safety. This is further illustrated by Luke in their first reflection:

> I took notes on everyone's design and presented three designs to the group that I thought were a good fit and implemented a small part of everyone's designs. There was a disagreement on a part of the design, and after listening to the other team member's reasoning I agreed, and we decided to add it to one of the designs. I think the team is good at making decisions because we all make sure to listen to each other and everyone has a voice.

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However, there was one instance where a team's vote overruled a team member's strongly advocated design. Although this team did reach a consensus through the vote, feelings of being overpowered and losing control were reported, "*There were no outward conflicts, but I feel there was tension, at least on my part. At the beginning I felt a little overpowered and like I had little control on the direction of our project*." This example, reported in Reflection 3 by Emma, highlights potential repercussions of high-task conflict, especially if there is a lack of compromise in the resolution.

4.2.2 | Absence of task conflict

Several students expressed the desire for their peers to share more of their opinions or ideas, signifying some groups had minimal task conflict. For some groups, this was just a challenge at the beginning of the semester as psychological safety was still being cultivated. This is captured well in the following focus group excerpt:

> There would be a prompt, and then I would say, "Okay, we should do this." Everyone would just look around and be like, "Okay." I didn't want it to be all my ideas. I honestly felt bad, but I just don't like sitting around and not talking. So, I would just always say things and let people .... I was like, give me something else. So, we ended up going, like in the beginning stages a lot, with how I wanted to do it. But as we became more comfortable, people were more open with saying, "Well, I think we should do it this way." And then people would be like, "Okay, yeah, for sure we can look into that." And so that really helped along the way, I think.

Although not a common occurrence, several participants describe having to make key project decisions without feedback from some of their team members despite their best efforts to solicit input. Process conflicts (e.g., absenteeism and lack of engagement) had an influence on task conflict. Our findings indicate the presence or lack of task conflict influences teams' ability to meaningfully navigate the design process and make design decisions. It is critical for teams to establish some psychological safety before having to make significant team decisions.

4.3 | Process conflict

Process conflict was prevalent in student teams, often triggered by student absences, ineffective communication, and procrastination on project tasks. Although some teams learned from minor process conflicts early in the semester and improved their collaboration, unresolved conflicts heightened stress levels and frustration among team members throughout the semester.

4.3.1 | Effects of absences during in-class team meetings

Process conflicts frequently surfaced from communication breakdowns, often catalyzed by team members' absences during critical in-class team meetings. In-class meetings were crucial for teams in project scoping, decision-making, and role assignment, highlighting the importance of attendance. Team member absences frequently led to challenges in coordinating tasks and making team decisions, as highlighted by Hannah in their first reflection:

> Some members do not communicate well such as not showing up for class when important decisions are made or when a meeting is scheduled outside of class time; members do not show up.

Absent team members were sometimes left out of important team decisions made during class, which could lead to further process conflicts. For instance, one team's absent member missed a discussion when a team converged on a project idea, leading to subsequent disagreements when they suggested alternative ideas upon return. Another common example discussed is absences negatively impacting a teams' ability to properly coordinate upcoming assignments or presentations. Absences, particularly unannounced, posed severe challenges, especially for teams unable to meet outside class hours. However, when these issues were addressed directly with the absent members, improvements in attendance were often observed. An example of this comes from Emily in her third reflection:

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> A major conflict was an issue regarding a team member not showing up for class time, where the majority of our planning and work took place. We put in our group chat that team members must show up for class time so we can collaborate and be on the same page about our project. We handled it pretty well and they started showing up to class more.

In some instances, open discussions with teammates revealed personal circumstances, such as surgery or bereavement, impacting attendance, emphasizing the importance of empathy in communication, and managing process conflicts effectively. Our data suggest absences on the team can lead to higher process conflict, which tend to negatively impact team effectiveness.

4.3.2 | Ineffective communication between class meetings

Many students reported difficulty coordinating tasks and making decisions through group chats, especially when this predicament was forced onto the team due to team member absences. Although the instructor informed students the need to update absentees on decisions and deadlines, texting was often deemed challenging and ineffective for communicating updates. Instances of delayed response times, one-word responses, or simply a lack of response from teammates impacted the team's ability to coordinate tasks outside of class. An example was written by Zoe in their first reflection:

> The class before our design review presentation, all but one of the group members were absent, leaving a majority of the team unaware of the presentation due next week. The member present in class texted us in great detail what we were assigned and what we had missed, yet it took the rest of us over 24 hours to respond back, including me. This delay in communication and setting a time to meet caused strain on the quality of our work.

Students generally describe the lack of timely responses, or responses at all, in group chats as disappointing and sometimes even strained their relationship with their teammates. In some cases, improved communication in group chats did occur once psychological safety was cultivated. This is expressed in a focus group:

> In the beginning for us, sometimes I'd ask a question in our group meeting, and it would just go unanswered for a whole week. Oh, it sucks. It happened three different times. And then after we actually got started to know each other, that's when everyone started communicating. So yeah, I don't know what that was about in the beginning, but it definitely got better over time.

Despite some success from other teams in task coordination via text, the need for in-person meetings was still strongly indicated as necessary and more efficient in coordinating and completing group tasks. This is highlighted by a student in a focus group:

> We dealt with a lot of issues with the CAD model, but we talked about it a lot on group chats on Discord, for example. Like he said, we had a Discord chat. Then we met up after class, we just sat somewhere and actually talked about it, and talked about all our ideas on how to improve it.

Despite recognizing the importance of in-person meetings, several teams struggled to coordinate these meetings for various reasons, such as team members living off campus, busy schedules, and simply team members needing to take the initiative to set up and implement a meeting time.

4.3.3 | Impact of procrastination

Procrastination emerged as a common hurdle, sometimes leading to delayed assignment submissions and diminished work quality. This was notably disruptive during in-class presentations, where last-minute preparations caused live miscommunications and teams not effectively addressing several presentation rubric criteria in the allotted time. In some

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cases, team members had to step in to complete the work of others who procrastinated to meet a deadline, as written in Max's second reflection:

> The only time I didn't enjoy working with a team member was a couple weeks ago when one team member was taking an extremely long time submitting their part of a group assignment, and I ended up having to do most of it for him. We got a good grade, but I lost some sleep.

In response to issues with procrastination, several teams introduced earlier internal deadlines and meeting times to rehearse presentations, and for subsequent tasks, the teams worked on a more organized preparation process. For some teams, implementing these strategies effectively mitigated the negative effects of procrastination, underscoring the significance of time management and structured coordination in team projects. Anthony describes this phenomenon in his second reflection:

> We procrastinated the proposal presentation creation a bit too much, and we were scrambling to get things together last minute. After our negative presentation, we came together as a group and all talked about what was wrong and how to fix it, which was also a very positive interaction.

Other teams struggled with procrastination throughout the semester. It was especially common during the prototyping phase at the end of the semester, with many students describing their prototypes coming together at the last minute. The interdependency of tasks amplified the negative impact of procrastination on final project deliverables. A prominent example in multiple teams was delays in completing certain prototyping tasks, such as 3D printing a part that other aspects of the project were contingent on. This is noted in a focus group:

> There were conflicts regarding the project themselves, such as having our parts printed on time, which they weren't. They were printed way later than we expected. So, we had to manage that and basically crammed the prototyping part of the project in a day or two.

Our data suggest procrastination among team members significantly affected teams, which impacted team effectiveness in meeting all the required deadlines to a high enough quality.

4.3.4 | Lack of participation and contribution from team members

Collaboration was challenged when certain team members consistently appeared disengaged and did not contribute their fair share of work to team tasks, despite other team members' best efforts and encouragement. Attendance issues, ineffective communication, and procrastination were influential process conflicts that impacted students' ability to participate and contribute to group work. Team members' repeated underperformance strained relationships over time, leading to relationship conflicts. Naomi illustrates this sentiment in her third reflection:

> The main conflict was how unprofessional [names removed] were in collaborating in the project. When trying to complete the brochure, none of them put in their fair share of work. X student and I put in the majority of the work, and it was taxing to remind them to do the bare minimum. We did not confront them on their lack of work after the project, so the conflict was never handled. I just wish they could have done their fair share of work.

Other notable process conflicts involved team members simply being distracted or working on other assignments during team meeting times. A particularly egregious example was shared in a focus group:

> We decided we wanted to have a Zoom meeting once .... We had to make a decision, and no one spoke on the phone call except me and the girl on the team .... It was the most strange experience, because we were like, "Hey guys, does anyone else have any input?" And for 30 minutes there's just silence. You're like, "Okay, this is so odd." Because you can see them and they're looking down at their phone. Someone was on

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> TikTok, and it was just flashing in their face, the different colored lights. We're like, "Okay." Oh my goodness.

Unsurprisingly, it can be uncomfortable for students when their team members are distracted, not engaged, or not paying attention during team meetings. This includes team members being distracted by other coursework as highlighted by Carlos' second reflection:

> One time that I particularly felt uncomfortable was when two of my group members spent the work time, focusing on another class material. I understood their position because some classes can get to be difficult, and one has to sometimes take time from some other tasks to get everything done.

Although this example was less egregious and easier to understand from the student's perspective, this scenario still negatively impacted students' comfort level. Distracted, unengaged team members can derail psychological safety and discourage healthy task conflict.

4.4 | Relationship conflict

4.4.1 | Persistent issues with process conflicts and unprofessional behavior

Although many teams managed process conflicts effectively, some faced recurring issues, most notably procrastination. Frustrations primarily arose when team members ceased communication before significant project milestones or failed to fulfill their responsibilities on time. Many students described tensions being especially high at the end of the semester due to procrastination on their project prototype. A representative example is shared by a student in a focus group:

> There was definitely a lot more stress and tension leading up towards the deadline than there probably could have been. So, what ended up happening a couple times was, while we'd be assembling our prototype, a small disagreement, since everyone was so tense, would escalate.

Egregious unprofessional behavior emerged on rare occasions, causing a significant rift in team dynamics. One such incident involved a team member consistently making politically incorrect and inappropriate comments, which the other team members found extremely distracting and discomforting. Another incident saw a team member lightly slapping another member in an attempt at humor, which resulted in discomfort and decreased enthusiasm of the recipient. Other team members noted a disruption in the team dynamic. This disruption is captured in a focus group:

> I don't exactly know the right word, how to describe him personally, but he had a little bit of anger issues later in the semester. But that kind of made the group dynamic a little, we were all hesitant to work together as much, and we kind of had some issues communicating throughout the semester just because I feel like a lot of the group members felt less engaged because of his interactions with us.

The course instructor addressed these incidents once notified. However, these examples underscore the fragility of psychological safety in a team setting, emphasizing how one team member can swiftly destabilize the group dynamic. Although instances of relationship conflict were less present in our data, when teams did experience relationship conflict, it significantly impacted team dynamics, psychological safety, and effectiveness.

4.5 | Conflict management processes

Three main conflict management strategies emerged from students' responses: avoiding confrontation, addressing conflict in the team, and reporting conflict to the instructor. For minor process conflicts (e.g., not responding to a text message), students discuss not being overly concerned. However, process conflicts that could impact graded group work (e.g., procrastination or unequal contributions) were often initially handled internally. If left unresolved, these issues were reported to the course instructor. Relationship conflicts, particularly toward the end of the semester, were typically

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reported to the course instructor. Notably, several students at the end of the semester discussed learning the importance of proactively documenting early process conflicts before they spiraled and openly communicating these issues with the instructor. This is captured in a focus group comment:

> I think my biggest thing is paper trail and communicating with instructors, even when signs start to get a little iffy because obviously you don't want to escalate.

Our data suggest teams have different conflict management approaches based on the significance of the conflict.

4.6 | Performance feedback

Although performance feedback was not a focus of data collection, feedback from both formal project checkpoints and informal discussions with instructors proved influential to teams, aiding decision-making and how teams address task conflict. For example, Kai shares in his third reflection:

> After the manager review, we realized how many issues and uncertainties we truly had, which led to the entire group working as hard as possible to form a solution to present. The communication of the team improved significantly, as we all were willing to admit when someone else had a better idea at the time.

Instructor feedback ranged from advising on project scope to mediating potential problems. Additionally, using CATME evaluations provided formative insights into the team dynamics, facilitating honest conversations with team members that could alleviate their stress. Several students remarked the course instructor was aware of their team issues and reassured team members who were contributing that their grades would reflect the effort they put into the project. This is illustrated by the following comment during the focus group discussion:

> I think our grades reflected the effort that we put into it. So, I think that also helped. And obviously our prototype was not up to the standards that we wanted it to be, but we got it done. So, it was hard, but we were very thankful that he [the instructor] was understanding, and he was aware of the situation the entire time, as opposed to last minute, "Oh, this is why our prototype is bad." We had proof early on that it was consistent problems, not like us failing at the last minute.

5 | DISCUSSION

This study aimed to explore psychological safety and conflict in first-year engineering student teams, tracing their evolution throughout a one-semester course. We uncovered nuanced insights into how students perceive the development of psychological safety and its interplay with various forms of conflict in a first-year engineering PBL class.

5.1 | Psychological safety in first-year engineering student teams

During the early stages of the project, psychological safety was not prevalent among most teams. Many students described early interactions as awkward and experienced hesitancy and uncertainty in idea-sharing. However, as team interactions progressed, positive affirmation and supportive environment contributed to development of psychological safety. This increased team members' confidence and willingness to share ideas, provide critical feedback, and display help-seeking behaviors. Our findings build on those of Cole, Marhefka, et al. (2022) and Cole, O'Connell, et al. (2022), who found psychological safety was associated with the number of quality ideas generated by first-year engineering teams. Our observations correspond with prior work on psychological safety, emphasizing the importance of early team interactions in creating a safe environment for open communication (Edmondson, 1999; Kahn, 1990). Further, we affirmed the critical role of positive reinforcement from team members and interpersonal risk-taking in nurturing psychological safety, aligning with the notion psychological safety is not static but a dynamic construct that can evolve over time (Edmondson & Lei, 2014).

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Our study also demonstrates there can be significant differences in the time it takes for teams to cultivate psychological safety. Some teams can break the ice in their first few team interactions. Meeting outside class significantly accelerated psychological safety as team members could get to know one another in informal contexts and develop rapport. In contrast, other teams needed weeks to begin feeling comfortable with one another in sharing ideas or addressing conflicts. Generally, teams appeared to cultivate psychological safety by the midpoint of the semester. Once psychological safety was cultivated, it generally supported collaboration for the remainder of the semester barring significant conflicts that were unresolved.

5.2 | Conflict in first-year engineering student teams

Our study suggests although task conflict was at least somewhat prevalent across most teams, students did not necessarily view it as conflict but rather as a difference in opinions or ideas. We found evidence of task conflict throughout the various stages of the design process and teams generally were able to reach consensus on decisions through open communication and collective problem-solving. Although not as prevalent, we also found evidence some teams had an absence of task conflict, demonstrated by student comments expressing frustration and longing for a more robust idea exchange. Interestingly, a situation where a team vote vetoed a strongly advocated design underscored the potential negative impacts of high-task conflict levels. Conversely, a lack of task conflict also posed challenges, suggesting first-year students find it difficult to navigate the opposite ends of the task conflict spectrum, particularly during early semester stages when psychological safety is still nascent. We also note perception of task conflict is subjective and can be viewed negatively depending on the relative importance of the disagreement, manageability, and interpretation of the conflict resolution (Jehn et al., 2008).

Process conflict was prevalent in our data, with team member absences and ineffective communication between class meetings significantly disrupting role assignments and decision-making. Procrastination also emerged as a commonly discussed process conflict that often impacted work quality and deadline adherence. Our findings thus align with those of Strenger et al. (2020), who found process conflict had the most significant influence on team member effectiveness, further confirming the prevalence of process conflict in this context.

Despite these challenges, many teams addressed process conflicts through improved communication and the implementation of effective project management strategies. It is plausible that managing process conflicts effectively led to improvements in team performance, as previously reported by Jehn and Mannix (2001). However, teams persistently grappling with process conflicts, particularly those procrastinating on final deliverables as the semester drew to a close, saw escalating relationship conflicts. This conflict not only heightened stress levels but seriously compromised psychological safety and the harmony of team dynamics. Our findings align with previous research that found unresolved, lingering process conflicts can lead to delays and inefficiencies (Jehn, 1997) and relationship conflicts lead to detrimental effects, including increased stress and inhibited cooperation (Jehn, 1995). Although these instances were rare, relationship conflicts emerged from other forms of unprofessional conduct, such as inappropriate comments leading to similar negative impacts.

For conflict management, students resorted to avoidance, internal conflict resolution, or reporting to the instructor as a last resort. Students discussed their conflict management processes differently depending on the type of conflict their team was facing and how the management process may have evolved as the conflict evolved. For example, process conflicts were often initially managed internally, but unresolved issues escalated into relationship conflicts, warranting instructor intervention. Interestingly, several students discuss a critical lesson learned is documenting early process conflicts and proactively communicating issues with instructors before they hamper group work. Overall, our study underscores the complex dynamics of conflict and critical role of effective conflict management strategies in these PBL teamwork contexts.

5.3 | The interplay of psychological safety and conflict

Throughout the semester, many teams collaborated effectively and nurtured psychological safety. However, unresolved process and relationship conflicts hindered psychological safety in some teams, particularly toward the semester's end. Our findings align and build off previous work (Beigpourian et al., 2019; Cole, Marhefka, et al., 2022; Cole, O'Connell, et al., 2022) by providing granularity on how psychological safety connects with the three conflict types (i.e., task,

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process, and relationship) in a first-year engineering PBL context. For example, our findings suggest psychological safety levels early in the semester influenced task conflict experiences, particularly during the project scoping and prototype planning phases.

Students highlight their reluctance to share ideas and their psychological need for affirmation early in the semester. Some students' minimal input or idea-sharing could be attributed to lacking psychological safety, although other factors such as apathy, lack of motivation, or other personal circumstances may have also played a role. As first-year students, many of them may not be ready to handle high levels of task conflict, especially in the early stages of group work while they establish psychological safety.

The ability of teams to effectively manage process conflicts affected psychological safety. For example, some students and teams felt comfortable in addressing process conflicts, such as procrastination or absences at team meetings, directly with the team. They could have meaningful discussions that led to resolutions, such as enforcing a team norm of having team assignments completed 48 h before a due date or texting teammates in advance if they could not attend a team meeting.

Other students and teams struggled to address process conflicts, perpetually struggling with the same issues over the semester. Failure to address a process conflict (e.g., texting team members and not receiving a response or asking for input from team members and consistently not receiving feedback) decreased psychological safety as these students and teams became more reluctant to try and address these process conflicts after prior failed attempts. This strained relationships over time and led to frustrating experiences and relationship conflicts. Although a rare occurrence, our findings also underscore the fragility of psychological safety in a team setting and how one team member can swiftly destabilize a team's psychological safety through unprofessional behavior, negatively impacting their ability to collaborate and the team's performance. Overall, our findings affirm the TCDM (O'Neill & McLarnon, 2018) applies to first-year engineering PBL contexts, emphasizing the critical role of psychological safety in effectively managing conflicts in student teams, thereby enhancing team performance and satisfaction.

Although not a focus of our study, our findings indicate performance feedback also impacted students' experience with psychological safety and conflict. Key checkpoints throughout the course in which the instructor reviewed work and provided students with regular formative feedback enabled students to assess their performance and progress. For high-performing teams, this feedback could reinforce student psychological safety. For other teams, this feedback served to illuminate process conflicts such as procrastination and the need for better project management or additional task conflict if a team needed to refine their solution further or improve the quality of their work. Additionally, some students noted the importance of the instructor monitoring peer evaluations and being aware of and addressing team issues. In some cases, this instructor intervention helped relieve the stress of students with reassurance their final project grade would reflect their effort in the course, even if limited contributions from other team members negatively impacted the quality of their final product.

5.4 | Recommendations for engineering instructional faculty

Based on our findings, we have identified general recommendations engineering instructional faculty should consider when designing and implementing engineering PBL and related courses that include team-based assignments and projects.

5.4.1 | Promote psychological safety from the outset

- *Early awareness*: Recognize at the semester's onset, many teams may not possess high levels of psychological safety, and individual students may require some affirmation when sharing their ideas.
- *Integrate icebreakers that promote psychological safety*: Incorporate content and activities that foster psychological safety in teams. For instance, icebreakers like team asset mapping (Stoddard & Pfeifer, 2018) encourage students to explore and appreciate each other's unique backgrounds, experiences, and strengths.
- *Encourage teams to meet outside of class*: Introduce an assignment that asks teams to convene outside of class in an informal setting and do an activity together of their choosing. This activity can accelerate the process of encouraging students to get to know one another and help cultivate psychological safety.
- *Introduce psychological safety*: Introduce the concept of psychological safety early in the classroom, referencing its significance through studies such as Google's internal study (Bergmann & Schaeppi, 2016), and encourage students to

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develop and share strategies that cultivate psychological safety. By bringing awareness to this terminology and emphasizing its importance, students can better appreciate and foster this crucial aspect of team dynamics.

5.4.2 | Strategically encourage moderate task conflict

- *Conflict framing*: Reiterate not all conflict is destructive by highlighting the value of task conflict for creativity, innovation, and collaboration. Some disagreement is necessary to build off the ideas of others in meaningful ways to develop effective solutions. We recommend reviewing the work of Ryan et al. (2023).
- *Avoid early group decisions*: Given the nascent stage of psychological safety, be cautious about demanding difficult group decisions early in the semester.
- *Promote task interdependence strategically*: Structure group projects to necessitate collaboration, discouraging the mere division of tasks among students. Have students complete individual work to ensure they are prepared to share ideas, concepts, and designs with their group members before having students complete a group task. This provides students the time and opportunity to generate their own ideas, which can promote more diverse perspectives and healthy task conflict.
- *Dynamic team norms*: While establishing team norms or charters is a common practice (Hagvall Svensson et al., 2024; Pertegal-Felices et al., 2019), instructors should encourage students to view these documents as evolving and have them revisit and update these documents periodically. Later in the semester, an additional assignment could involve having teams debrief on how effectively the teams are adhering to the previously established team norms and make updates.
- *Regular debriefs*: Facilitate in-class team debriefs of discussions that invite students to discuss and learn from recent team experiences after key course milestones. This could include discussions on common process conflicts identified in this study, ensuring teams understand they can expect to experience these conflicts during the semester. Teams should apply conflict resolution strategies to address them. Debriefs have been demonstrated to improve team performance in various contexts (Tannenbaum & Cerasoli, 2013).
- *Documentation*: Require teams to submit weekly meeting notes and revolving action item list with assigned tasks and task status updates. Encourage teams to address process conflicts directly with the team and document these discussions in their weekly minutes to be visible and actionable.

5.4.3 | Monitor teams to address potential relationship conflicts proactively

- *Conflict monitoring*: Regularly observe team dynamics, particularly indicators of teams lacking task conflict, experiencing process conflicts, and having unprofessional conduct that can trigger relationship conflicts.
- *Peer evaluations*: Introduce recurring peer evaluations throughout the semester, ensuring students become more comfortable and transparent with feedback. Provide basic training to students on how to provide peer feedback and assess the quality of the feedback provided, not just whether the students completed the evaluation.
- *Open communication*: Instructors should assure students of their availability to address unresolved team conflicts and provide a safe setting (e.g., office hours) where they can discuss issues openly and honestly.
- *Holistic evaluation*: Beyond the final project deliverable, assess the collaborative process itself through periodic assignments as already discussed (e.g., weekly meeting notes, and peer evaluations). Consider directly evaluating teamwork and communication through performance-based grading or even consider a professionalism grade as a component of the final course grade to reward effective collaboration and penalize lack of contribution or inappropriate conduct.

By adopting these recommendations, instructors can better facilitate a productive, harmonious, and effective learning environment for first-year engineering students with consideration for the impacts of psychological safety and conflict on team dynamics.

5.5 | Limitations

Our primary limitations stem from the specific course, institutional context, and study design. This study focuses on two sections of a specific course sequence, influenced by one instructor's pedagogical strategies. It is crucial to note the possible

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influence of the instructor's approach on the organizational environment, team dynamics, and development of psychological safety. As this course was an off-cycle semester course section, this class had more nontraditional students than on-cycle semester course sections. Student demographics in our study only partially mirror the larger university engineering population. For example, 37% of participants indicated their racial/ethnic identity as Asian, but only 15% of the engineering population at the university identifies as Asian (Virginia Tech, 2024). The gender breakdown in this study was representative of the engineering population at the university, with 20% of our participants identified as women.

Our study design and sample of first-year engineering teams did not enable us to explore the role of team composition and demographics on psychological safety and conflict. Although our data were limited to two classes, several notable examples of how team composition influenced psychological safety emerged in our data. For instance, a student from an all-male team experienced higher psychological safety early in the group project, which he attributed to their homogenous team composition. Another example involved mixed-gender teams forming gender-based subgroups, which negatively impacted the team's psychological safety.

These examples appear to align with the findings of Wei and Ohland (2022), who found all-male teams had significantly higher psychological safety than teams with mixed-gender students and confirmed other findings that suggest diversity in teams can introduce obstacles to a team's psychological safety (Jimenez-Useche et al., 2015) and amplify potential conflicts (Neumeyer & Santos, 2020). However, we note diversity in teams can be a potent catalyst for learning and innovation and contribute to healthy task conflict through the exchange of diverse perspectives (De Dreu & Beersma, 2005; Desivilya et al., 2010). The importance of establishing inclusion and belonging, especially in diverse teams, is echoed in the findings of Rodríguez-Simmonds et al. (2023) in their study on first-year engineering teams.

Students completed CATME surveys throughout the semester, which includes scales on psychological safety and conflict, and are not included in our findings. We did analyze this quantitative data and identified several correlations aligned with the findings presented in this paper. However, the data did not fully align with the findings presented here and it was not normal, limiting validity of these findings and our ability to analyze this data. We opted to prioritize reporting the qualitative data in detail in this paper.

5.6 | Future work

An intriguing dimension for future work that can build off of our study is the influence of team composition on the formation of psychological safety and management of conflict. A comprehensive exploration of how team composition factors (e.g., gender, English fluency, engineering skills, personality characteristics) influence early formation and development of psychological safety can help shed further light on how best to foster psychological safety in diverse teams. This can include the instructor's consideration of these factors when forming and mentoring teams to promote inclusion and healthy team practices. Additionally, further consideration for contextual features warrants further exploration such as the organizational environment and team design (as outlined in the TCDM), such as how course design, phases of the design process, and team characteristics influence conflict. Further, identifying the optimal amount of task conflict that educators should encourage, and at which stages, could enhance pedagogical outcomes.

Moreover, validating classroom interventions that help student teams rapidly foster psychological safety and adopt effective conflict management strategies is paramount. By leveraging the TCDM, we can investigate how these interventions impact the intricate balance between psychological safety and conflict management. The TCDM's team performance feedback loop component is another area for future exploration. Although not a focus of our study, understanding how the course structure and nature of feedback mechanisms influence team psychological safety and conflict dynamics can refine our educational practices. This is especially true of the peer feedback process. Previous research has highlighted how students may be reluctant to provide critical feedback to their peers due to social discomfort and potential negative impacts on team dynamics (Mayfield & Tombaugh, 2019).

Lastly, contrasting the experiences of first-year engineering students in a PBL setting with those of capstone student teams offers an insightful perspective. With their additional experience, capstone students may differ significantly in their approach to cultivating psychological safety and managing conflict. Probing questions similar to this study that explore their experiences establishing psychological safety, receptiveness to task conflict, and encounters with process and relationship conflicts could be illuminating. However, to ensure a meaningful comparison, it is crucial to account for differences in contextual factors like course design and team formation and how these may impact students' approaches to conflict management (Schibelius et al., 2023). By venturing into these research domains, we can better understand team dynamics in educational settings, paving the way for more informed curriculum design and teaching approaches.

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6 | CONCLUSION

Effective teamwork underpins engineering work with psychological safety and conflict management as critical components. Despite their importance, a notable research gap exists regarding students' experiences with these elements in engineering PBL environments. Furthermore, the literature scarcely addresses faculty approaches that foster these teamwork components, particularly in nurturing psychological safety. This study bridges this knowledge gap, spotlighting students' perceptions in a first-year engineering PBL context.

Our findings revealed at the beginning of the semester, student teams had not established a strong sense of psychological safety, manifesting in students' reluctance to voice ideas due to a fear of negative judgments. As the semester progressed, consistent positive feedback from peers and informal interactions outside of the classroom helped foster a psychologically safe environment and encouraged idea-sharing. Throughout the design process, the presence and lack of task conflict played a pivotal role in team outcomes. Although some teams appeared to effectively discuss and critique ideas, potentially leading to higher innovation, others expressed frustration at a perceived lack of task conflict, desiring more collaborative ideation and decision-making.

Interestingly, several students describe perceiving minor disagreements in ideas as nonconflicts. However, process conflicts, commonly associated with disagreements on project procedures and responsibilities, were ubiquitous across many teams. Root causes included absenteeism, unequal contributions, poor communication, procrastination, and distracted team members. Encouragingly, many teams adeptly navigated these challenges, devising robust project management and communication strategies. Conversely, other groups grappled with unresolved or late-emerging conflicts, subsequently eroding psychological safety and spawning relationship conflicts, which amplified stress and intensified frustrations among team members.

Our study underscores the intertwined nature of psychological safety and conflict in shaping the first-year design experience in student teams. Our research emphasizes the need to leverage these insights to inform pedagogical approaches that support students' ability to cultivate and maintain psychological safety in their teams and effectively manage conflicts that inevitably emerge (whether recognized by students as conflicts or not). We provided an initial list of specific recommendations for educators and highlight areas of related future work.

Building an inclusive culture and catalyzing innovation in engineering hinges on effective teamwork rooted in psychological safety and effective conflict management. Faculty play a unique and critical role in this ecosystem, acting as linchpins who can set the tone for team dynamics, shaping their students' academic experiences and career preparation. Every classroom, academic-related experience, or interaction is a potential pivot for growth in a student's journey. We urge faculty to recognize their pivotal role as instructors and architects of team dynamics and stewards in fostering psychologically safe environments. Educators must recognize this responsibility, actively stepping forward to equip students with the skills to excel in team-based endeavors and, in doing so, redefine the contours of engineering's future.

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APPENDIX A

REFLECTION AND FOCUS GROUP QUESTIONS

Instructions: The purpose of this assignment is to reflect on your team experiences thus far and consider ways to improve your team's overall effectiveness and your own approach to team interactions. The questions in this reflection will have a similar format to the STAR technique, which can be useful for interviews for future engineering jobs that often include behavioral questions that start with "Tell me about a time ...." This technique can prepare you to answer behavioral questions like these by bringing the interviewer through a story. The method is as follows:

- Situation—Describe the context of the situation and the challenge you faced.
- Task—State your role or responsibility in the situation.
- Action—Explain the steps you took to overcome the challenge.
- Result—Describe the outcome that occurred based on your actions.

Using the STAR format, write four to six sentences that address each of the prompts and sub-questions.

Reflection 1

- In a paragraph (four to six sentences) describe a representative example of how your group delegated tasks for a group assignment. Was your approach effective? Are there ways your team could improve in delegating group work?
- In a paragraph (four to six sentences) describe a time your group had to make an important or difficult project-related decision so far this semester. Were all of the perspectives of the group members considered? Why or why not? Were there any disagreements or conflicting opinions? If so, how were the disagreements among team members handled? Are there ways the team could improve in how it makes important team decisions?
- In a paragraph (four to six sentences) describe a time when your group did not have effective communication so far this semester. How are you making sure these issues do not happen again? If you haven't had this happen, how do you plan on dealing with it in the future?
- In a paragraph (four to six sentences), describe the process your team followed to develop and maintain your team community and norms. To what extent do you feel comfortable sharing your thoughts and ideas during team interactions? To what extent do you think your other team members feel comfortable sharing their thoughts and ideas during team interactions?

Reflection 2

- In a paragraph (four to six sentences), explain your role and responsibilities leading into the final prototyping and presentation phase of the semester. How did the team decide on the roles and responsibilities for each team member? Do you think the team's approach in delegating roles and responsibilities has been effective?
- In a paragraph (four to six sentences), tell me about the most important decision your team had to make this semester. What did that discussion look like? Do you think the team's approach to making the final decision was effective?
- In a paragraph (four to six sentences), share a time you really enjoyed collaborating with a team member and a time you didn't enjoy working with a team member this semester. To what extent have you been able to effectively manage your relationships with other team members? What was the outcome of each of those scenarios? Were your goals met?
- In a paragraph (four to six sentences), explain how your ideas and thoughts have been perceived by your group. How has your comfort level changed throughout the semester in terms of your willingness to share your thoughts and ideas?

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Reflection 3

- How well do you think your team collaborated this semester? Explain your answer. What were the primary benefits of collaborating with others to complete the project? What were the primary challenges or barriers in collaborating with others to complete the project? These can be personal, related to other team members, or other circumstantial reasons.
- Were there any conflicts that you and/or your team had to manage this semester? If yes, what was the conflict? Why did it happen? How did your team handle that conflict? Is there anything you wish you or others did differently? If no, how was your team able to prevent conflict during the semester? Did the conflict management workshop (if you were present for this class) provide value in helping you handle conflict during the semester?
- To what extent did your team dynamics evolve or change over the course of the semester? Was there any difference between how your team functioned at the beginning of the semester compared with the middle and end of the semester? Explain your answer.
- What will you remember or take away from your teamwork and overall class experiences this semester? Do you believe you further developed and improved your teamwork skills this semester? This could be relevant to communication, managing conflict, managing a project, learning how to work with different people, and so forth. Explain your answer. Was there any particular activity and/or aspect of the course design that helped you improve your teamwork skills?

Focus group

1. What were your overall impressions working with your team this semester?
   a. Was it what you expected? What did you like? What did you dislike?
2. How well did your team collaborate?
   a. What were the primary benefits of collaborating with others to complete the project?
   b. What were the primary challenges or barriers in collaborating with others to complete the project? These can be personal, related to other team members, or other circumstantial reasons.
3. To what extent did your team dynamics evolve or change over the course of the semester?
   a. Was there any difference between how your team functioned at the beginning of the semester compared with the middle and end of the semester? Explain your answer. What contributed to any changes?
4. Were there any notable conflicts that your team had to manage this semester?
   a. What was the conflict? Why did it happen? How did your team handle that conflict? Is there anything you wish you or others did differently?
5. What will you remember or take away from your teamwork experiences this semester? Do you believe you further developed and improved your teamwork skills this semester? This could be relevant to communication, managing conflict, managing a project, learning how to work with different people, and so forth.
   a. Explain your answer and specifically how you think your teamwork skills improved.
6. Was there any particular activity and/or aspect of the course design that helped you improve your teamwork skills?
7. Does anybody have anything else they would like to share that they were not able to?

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APPENDIX B

CODEBOOK FOR PSYCHOLOGICAL SAFETY, TASK CONFLICT, PROCESS CONFLICT, AND RELATIONSHIP CONFLICT

Summary of psychological safety themes and example student quotes:

Theme	Description	Example excerpt (*Note*: Some excerpts were condensed for brevity and flow)
Improvements in psychological safety and open communication	Over the semester, psychological safety in student teams grew, starting from initial awkwardness and uncertainty toward more confident idea-sharing, facilitated by consistent positive exchanges and the development of an open, affirming environment.	"*I was a bit quiet in the beginning stages of the project, but I suggested a few ideas to my team which they seemed to really like. As a result, I felt encouraged to share more of my ideas and feedback with my team members which they have taken with a positive attitude and are appreciative of.*" *– Chin [Asian Male], Reflection 3*
Evolution of psychological safety	The evolution of psychological safety in teams varies, with some quickly establishing comfort and open communication early on, but others take longer, often needing to overcome initial social boundaries, but once established, this safety tends to stabilize the team dynamic for the semester.	"*My team, for example, was a melting pot. It was cool. Definitely was a learning curve at the beginning, just trying to meet everyone and understand everyone. But as the time went on, everything worked itself out and we were able to work pretty well together. But getting the teams I guess was a little weird.*" *– Focus group*
Cultivating psychological safety outside of class	Meeting outside of class in informal settings significantly enhances the cultivation of psychological safety in teams, fostering friendliness and ease of communication.	"*We definitely got more comfortable around each other. After the first couple of weeks, we started really opening up, and we even hung out a couple times just for fun, which I think really helped team morale.*" *– Focus group*
Unresolved conflicts influence psychological safety	Lingering process or relationship conflicts, especially toward the end of the semester, strained team dynamics and undermined psychological safety, with unresolved tensions and frustrations significantly altering group interactions.	"*The dynamic of the team changed greatly due to frustration toward other members and its pace grew exponentially, in an unhealthy manner.*" *– Arjun [Asian Male], Reflection 4*

Summary of task conflict themes and example student quotes:

Theme	Description	Example excerpt (*Note*: Some excerpts were condensed for brevity and flow)
Task conflict		
Evidence of task conflict	Task conflict in teams, often manifesting as differences in opinions rather than outright disagreements, was managed effectively through open communication and mutual respect, underpinned by psychological safety, leading to constructive decision-making and design consensus.	"*When we were trying to find a problem to solve. We made a document, where everyone got to write their ideas. We even let it for several days so everyone can think and add whatever they like. Then, we voted for the best one. So yes, all the perspectives were considered. There weren't any disagreements.*" *– Mariam [Middle Eastern Female], Reflection 2*
Absence of task conflict	Some teams displayed minimal task conflict, with students expressing a desire for more peer input. The lack of contribution led to certain individuals making team decisions with little peer feedback, potentially hindering collective ideation.	"*We had to choose between three different options: implementing our prototype through the use of a motion, sound, or pressure sensor. I attempted to consider all the perspectives of the group members. However, I did not receive much input from the others. I choose to implement a sound sensor.*" *– Ellie [White Female], Reflection 2*

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Summary of process conflict themes and example student quotes:

Process conflict		
Effects of absences during in-class team meetings	Absences at key meetings without prior notice triggered process conflicts by disrupting role assignment and decision-making, though empathetic, direct discussion between team members addressing the issue often improved the situation.	"*A major conflict was an issue regarding a team member not showing up for class time, where the majority of our planning and work took place. We put in our group chat that team members must show up for class time so we can collaborate and be on the same page about our project. We handled it pretty well and they started showing up to class more*." *– Emily [White Female], Reflection 4*
Ineffective communication between class meetings	Teams experienced communication issues between team meetings and struggled with coordinating tasks and making team decisions via texting.	"*I can tell you straight up communication in our group has been a problem. I've always enjoyed working with the group in person, and we have no problems there. However, we fell extremely short when it comes to communicating outside of class*." *– David [Asian/White Male], Reflection 3*
Impact of procrastination	Procrastination, often leading to delayed submissions and reduced work quality, was magnified during interdependent tasks such as the prototyping phase. In response, some teams mitigated these issues by setting earlier internal deadlines and improved preparation processes.	"*We procrastinated its [proposal presentation] creation a bit too much and we were scrambling to get things together last minute. After our negative presentation, we came together as a group and all talked about what was wrong and how to fix it, which was also a very positive interaction*." *– Anthony [White Male], Reflection 3*
Lack of participation and engagement from team members	Team collaboration suffers when members are disengaged or fail to contribute equitably, leading to strained relationships and conflicts arising from issues like attendance, poor communication, procrastination, and distraction during meetings.	"*The others did absolutely nothing to help me. I gave them a heads-up and texted several times in the group chat, but no one else helped me complete it. My relationship with the others is strained on my side, as I am disappointed by their lack of investment in this project*." *– Lydia [White Female], Reflection 3*

Summary of relationship conflict themes and example student quotes:

Relationship conflict		
Persistent issues with process conflicts and unprofessional behavior	Continuous struggles with process conflicts, particularly procrastination and failure to meet commitments, strained relationships and magnified frustrations. Rare instances of unprofessional conduct, including inappropriate comments, had a profound negative impact on team dynamics and psychological safety.	"*The main conflict was how unprofessional XXX, XXX, and XXX were in collaborating in the project. When trying to complete the brochure, none of them put in their fair share of work. It was only after multiple times of asking them to turn in their work that they finished essentially one slide each. XXX and I put in the majority of the work, and it was taxing to remind them to do the bare minimum*." *– Naomi [Asian/White Female], Reflection 4*

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